COMET HACKATHON | 5G & 6G INNOVATION

DISASTER-RESPONSE WI-FI BEACON LOCATOR

AI + IoT + Edge Computing + Drone-Assisted Search & Rescue

When networks fail, every signal can guide a search.



TEAM GEO NEXUS

Prototype concept • Survivor search-zone prioritization

SOLUTION WORKFLOW

From wireless signals to real lives

AI + IoT + Edge Computing + Drone-Assisted Search & Rescue

Detect → Analyze → Locate → Save

1 SENSE

Collect Real-time & Multi-Source Data

- Scan permitted Wi-Fi transmissions & capture signal strength (RSSI)
- Read environmental sensors (temperature, humidity, motion, etc.)
- Collect drone observations (aerial images & video)

2 FUSE

Integrate & Build a Unified View

- Combine signal measurements (RSSI, device info)
- Add sensor readings (environmental + power)
- Include GPS coordinates (from ground & drone)
- Merge aerial observations (images + video)

3 PRIORITIZE

ML-Based Search Zone Estimation

- Analyze combined data using Machine Learning (Random Forest / Classification)
- Estimate likely search zones and confidence scores
- Rank areas by urgency and risk level

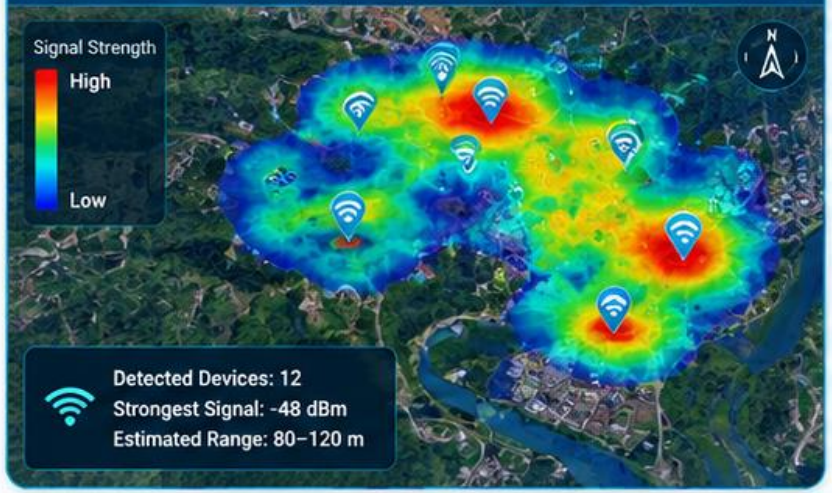
4 RESPOND

Guide & Coordinate Rescue Operations

- Display priority zones on live dashboard
- Guide rescue teams (with route suggestions)
- Dispatch drone for further inspection

Wi-Fi Signal Heatmap

Visualizing detected wireless signals and their strength



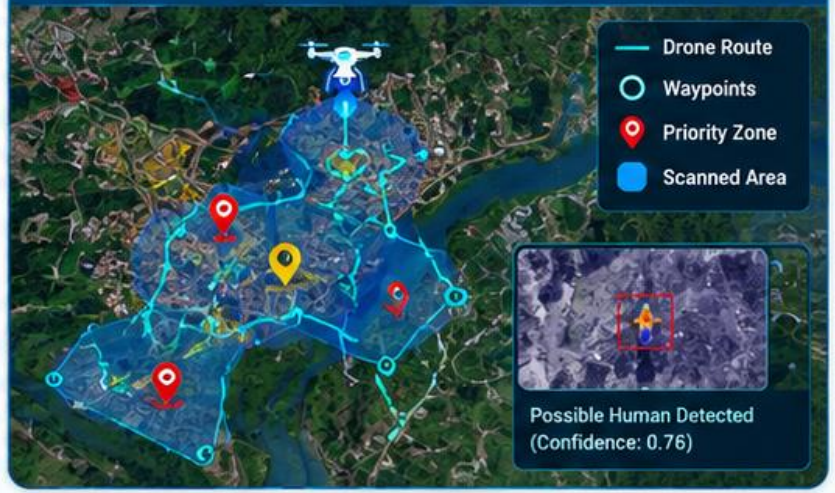
Rescue Priority Map

AI-driven search zones with confidence levels



Drone Route & Search Plan

Autonomous/Operator-Guided Inspection Path



FASTER RESPONSE . SAFER COMMUNITIES . SAVE LIVES

TECHNICAL APPROACH

Drone-Assisted Search & Rescue | IoT + Wi-Fi Sensing + Machine Learning

Geo Nexus SVIST

DATA SOURCES

Wi-Fi signals
Permitted transmissions • RSSI • timestamps

IoT sensors
Environmental readings • device status

GPS & ground nodes
Receiver coordinates • observation time

Drone observations
Aerial images • video • optional thermal

DATA PROCESSING

1. Collect
2. Clean & validate
3. Synchronize time + coordinates
4. Extract features

EDGE / COMMAND PLATFORM
Local buffering when offline • secure data transfer when available • operator dashboard

FEATURE ENGINEERING

- RSSI patterns over time
- Signal observation count
- Sensor and device status
- GPS / drone position
- Visual scene cues
- Signal and location uncertainty

ML SEARCH-PRIORITY MODEL

- Rank candidate zones
- Confidence + uncertainty
- Explainable baseline: Random Forest
- Validate using detection rate, false alarms, localization error, latency

DRONE-ASSISTED SEARCH & RESCUE



1 SURVEY
Inspect inaccessible or unsafe terrain

2 OBSERVE
Capture aerial images/video; optional thermal sensing if equipped

3 REPORT
Send observations to command dashboard for human review

REPORT
GIT HUB
(click here)

HUMAN RESCUE
Flag possible person / distress cues for verification

ANIMAL RESCUE
Identify possible trapped or stranded animals for verification

COMMAND DASHBOARD



Decision support only: ML ranks areas to investigate; it does not confirm survivor presence. Routes and flight operations remain operator-approved and subject to local safety rules.

DRONE-ASSISTED SEARCH & RESCUE

Aerial intelligence • Edge AI • Human & animal rescue

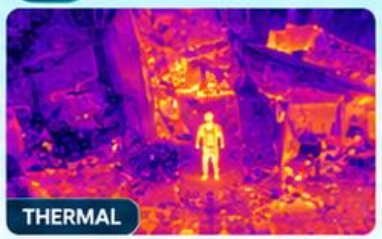
MISSION



- Inspect inaccessible or hazardous areas
- Capture aerial RGB imagery and thermal observations where equipped
- Follow operator-approved search routes
- Stream or store observations for the command dashboard

DETECT & PRIORITIZE

THERMAL + VISUAL





- Flag possible people or animals for human review

DRONE-MOUNTED WI-FI SCANNING





- Combine compatible wireless observations with drone position to refine search zones

⚠ *Thermal signatures and Wi-Fi signals are indicators, not proof of a survivor.*

RESCUE WORKFLOW

1. Survey zone
2. Detect & map clues
3. Rank search areas
4. Rescuers verify

COMMAND DASHBOARD







SAFE, PRACTICAL EXTENSIONS

HUMAN + ANIMAL RESCUE
Separate possible-human, possible-animal and unverified detections for review.

EMERGENCY SUPPLY RELAY
Future extension: deliver a lightweight kit, radio or marker only after human verification and route-safety checks.

Requires compatible thermal/visual payloads and validated wireless receiver; capabilities depend on hardware and testing.

HillGuard

Early Warning · Faster Response · Safer Lives

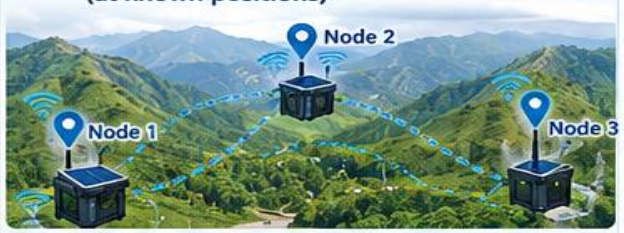
Smart Wireless Localization for Disaster Rescue

Using Wi-Fi signals, multi-source data and intelligent analytics to find the lost, even in the toughest terrains.

Team Geo Nexus

Mapping Today | Saving Tomorrow

1 Deploy Multiple Scanning Receivers (at known positions)



- Place scanning nodes at safe, known locations.
- Each node logs observations and relays data to the command system when possible.
- Multiple nodes help estimate where the signal was observed most strongly.

2 Record Measurements

Time	Receiver Location	Signal Strength
10:24:15	26.142°N, 92.847°E	-62 dBm
10:27:33	26.145°N, 92.851°E	-68 dBm
10:31:02	26.138°N, 92.840°E	-71 dBm



- Record the time, receiver coordinates and signal strength for observable transmissions.

3 Compare Measurements



Node 1 -62 dBm

Node 2 -68 dBm

Node 3 -71 dBm

- Compare measurements from different receivers or different positions of a moving receiver.

4 Estimate a Likely Region



High Confidence
Medium
Low
Uncertainty Area

N

- Estimate a likely region, not a guaranteed exact coordinate.
- Represent uncertainty on the map.

5 Send Rescue Team or Drone



- Send a rescue team or drone to investigate and verify the location.

6 Offline-First Disaster Mode

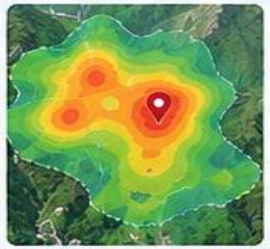


Store locally during outage

Sync when connection returns

- Keep scanning and recording locally when the internet is unavailable.
- Synchronize data when the link returns. This is more practical than assuming a constant cloud connection.

7 Search-Priority Heatmap

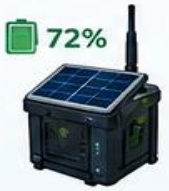


Priority Zones
- Very High
- High
- Medium
- Low

Why this area?
- ✓ Repeated Wi-Fi signals
- ✓ Drone detection
- ✓ Terrain context
- ✓ Historical data

- Display priority zones based on wireless evidence, repeated observations, drone detections and environmental context.
- Let the operator see why an area was ranked highly.

8 Power-Aware Operation



Battery	72%
CPU	Normal
Storage	Normal
Signal	Good

- Record battery level and system health for each portable node.
- Prioritize low-power scanning and alert the team before a node stops collecting useful information.

9 Privacy-Preserving Detection

Only store:
- ✓ Signal strength
- ✓ Timestamp
- ✓ Location (approx.)

Do NOT collect personal device data or identify individuals.

- Store only the signal features required for localization.
- Avoid collecting personal device data unnecessarily.
- Use short retention periods.

10 Human and Animal Rescue Classification



Manual Review
- ✓ Reduce false alerts
- ✓ Don't miss animals
- ✓ Improve accuracy

- Use drone imagery to flag possible humans or animals separately.
- Send ambiguous cases for manual review.

11 Risk-Aware Search Routing



Safe Route
Blocked Path
Hazard Zone

Human Approval Required

Known Hazards | Blocked Paths | Accessibility

- Rank safe, accessible search routes using known hazards and blocked paths.
- Require human approval before the drone or a rescue team is sent into a dangerous area.

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Arduino UNO Q + 5G/6G

The Hardware Backbone of Our Solution

Smart Sensing | Edge AI | Connected Rescue

Team Geo Nexus

Mapping Today | Saving Tomorrow

Why Arduino UNO Q?

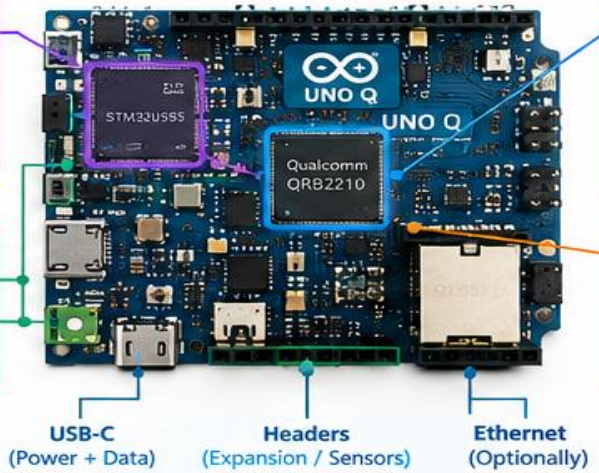
- High-performance, all-in-one development board
- Linux-based + real-time microcontroller
- Built-in Wi-Fi 5 & Bluetooth 5.1
- Supports AI/ML, sensor fusion, and edge computing
- Flexible for 5G (via USB module) & future 6G research

Designed for **COMET Hackathon** and ideal for real-world prototyping.

Arduino UNO Q – All-in-One Dev Board

STM32U585 Microcontroller
- Real-time sensor data acquisition
- Peripheral control
- GPIO & I/O tasks

Sensors & I/O
- IMU, temperature, pressure, etc.
- Camera / expansion headers



Qualcomm Dragonwing QRB2210 Processor
- Linux-based application
- Python processing
- ML inference
- Data fusion
- Local storage
- Dashboard / backend

Communication
- Wi-Fi 5 (dual-band)
- Bluetooth 5.1
- USB 5G module (optional)
- 5G/6G (future research)

Simple Responsibility Split

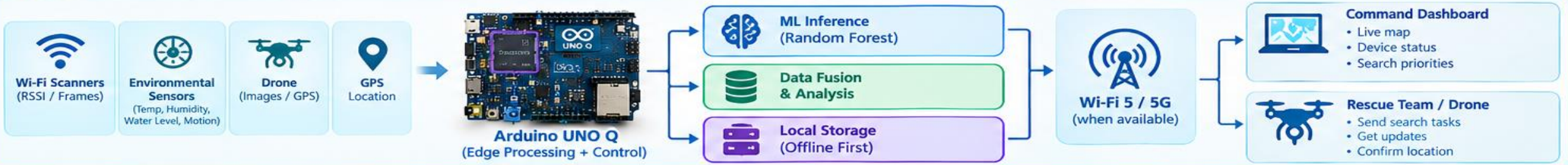
Qualcomm Dragonwing QRB2210
- Linux application
- Python processing & ML inference
- Data fusion, storage & dashboard/backend

STM32U585
- Real-time sensor acquisition
- Peripheral control
- Hardware tasks

Communication Layer
- Built-in Wi-Fi 5 & Bluetooth 5.1
- Optional 5G module (USB)
- 6G (future compatibility / research)

Rescue Interface
- Local map & search zones
- Scanner status & last observation
- Confidence score & drone status
- Operator-approved search tasks

System Architecture & Data Flow



How It Works – End to End Flow

1. **Collect Data** — Wi-Fi signals, sensor readings, GPS, drone images.
2. **Process on UNO Q** — Run ML model (Random Forest), fuse data, estimate priority zones.
3. **Show on Dashboard** — Map, confidence score, scanner & drone status.
4. **Coordinate Rescue** — Send priority zones to teams or drone (when available).
5. **Verify & Update** — Confirm location, update model with new data.
6. **Save & Improve** — Store results (offline), learn for better accuracy.

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System Architecture & Machine Learning Pipeline

From Wireless Signals to Real-World Rescue

How our App is different

Team Geo Nexus

Mapping Today | Saving Tomorrow

1 System Architecture
7-Stage Pipeline

01 IoT Sensing Layer

Wi-Fi Scanner
- Collect wireless frames
- Signal strength

Location Reference
- GPS / known position

Environmental Sensors
- Temp, humidity, water level, motion

Drone Observations
- Images, thermal (if available)
- Aerial coordinates

Sensor readings + signal features + location metadata

02 Edge ML on Arduino UNO Q



Feature Extraction
- Signal strength over time
- Detection frequency
- Multi-position comparison

Search-Priority Estimation
- Identify high-priority zones
- Show supporting evidence
- Estimate uncertainty

Python ML + Sensor Fusion on UNO Q

03 Rescue Action

Live Search Map
- Detected sources
- Scanner positions
- Estimated zones
- Timestamps & confidence

Rescue Coordination
- Send to command dashboard
- Relay to field teams
- Require human approval for risky areas

Alerts + Map + Field Teams + Drones

Which Machine Learning Model?
Start simple. Train, test, explain, and demonstrate.

ML Method	What it does in our project	Priority
Random Forest	Classifies device present/absent or high/low search priority using labelled features (RSSI, count, location, etc.).	Best starting point
Regression Model	Estimates distance or location-related values from calibrated signal features.	Optional, requires real calibration
Computer Vision	Identifies people, animals or visible hazards in drone images.	Advanced extension
Sensor Anomaly Detection	Flags unusual changes in water level, motion or environmental readings.	Optional extension

A classifier trained on a controlled test setup does not prove it can detect actual trapped survivors. Include negative cases, background devices, different distances, obstructions and changing conditions.

Prototype Validation Metrics (Targets)

- **Detection rate** ≥ 90%
- **False-alarm rate** ≤ 10%
- **Localization error** ≤ 50 m
- **Time to identify a test device** ≤ 2 minutes
- **% of data processed offline** ≥ 70%

Validation setup
- Controlled test area
- Multiple distances & obstructions
- Real device + background devices

More System Features

1 Portable Rescue Beacon Nodes
Place scanning nodes at safe, known locations. Each node logs observations and relays data to the command system. Multiple nodes improve accuracy.

2 Offline-First Disaster Mode
Keep scanning and recording locally when internet is unavailable. Synchronize data when the link returns.

3 Search-Priority Heatmap
Priority zones based on wireless evidence, repeated observations, drone detections and environmental context. Shows why an area is ranked highly.

4 Power-Aware Operation
Track battery & system health for each node. Use low-power scanning and alert the team before a node stops collecting useful information.

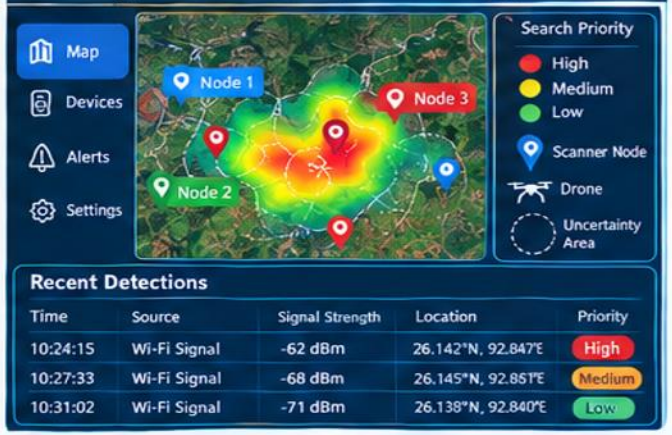
5 Privacy-Preserving Detection
Store only signal features required for localization. Avoid collecting personal device data. Use short retention periods.

6 Human and Animal Rescue Classification
Use drone imagery to flag possible humans or animals separately. Send ambiguous cases for manual review.

7 Risk-Aware Search Routing
Rank safe, accessible search routes using known hazards and blocked paths. Require human approval before the drone or rescue team is sent into a dangerous area.



Prototype Dashboard (HillGuard App)



Recent Detections

Time	Source	Signal Strength	Location	Priority
10:24:15	Wi-Fi Signal	-62 dBm	26.142°N, 92.847°E	High
10:27:33	Wi-Fi Signal	-68 dBm	26.145°N, 92.851°E	Medium
10:31:02	Wi-Fi Signal	-71 dBm	26.138°N, 92.840°E	Low

Measured Benefits & Deployment Scenarios

Measurable Benefits
- Faster search & rescue response
- Improved localization accuracy
- Better use of resources
- Reduced risks to rescue teams
- Lower false alarms (with ML)

Feasibility
- Uses available hardware (UNO Q, sensors, drones)
- Open-source ML (scikit-learn)
- Works offline
- Scalable and modular

Limitations
- Signal can be weak or inconsistent
- Weather & terrain affect results
- Not a guaranteed exact location
- Requires calibration & training data

Deployment Scenarios
- Mountainous regions & remote areas
- Forests, landslides, floods
- Disaster relief & humanitarian missions
- Integration with existing response systems

Business Model
Click Here

Sustainable. Scalable. Impactful.

Product Sales
- Hardware kits (scanners + sensors)
- Software license

Subscription Services
- Cloud sync, map analytics, alerts & support

Government & Disaster Agencies
- Public safety, forest departments,
- NDMA / SDRF

Enterprise & NGO Partnerships
- Research, field deployment, custom solutions

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RESEARCH & REFERENCES

Team Geo Nexus

Mapping Today | Saving Tomorrow

PROJECT RESOURCES

Project Report
Full research report and methodology.

Research PDF
Research paper and data analysis.

GitHub Repository
Source code and technical details.

RESEARCH & TECHNICAL REFERENCES

1. **Wi-Fi RSSI & CSI Localization**
   Abuhoureyah et al. (2023), IET Wireless Sensor Systems.
   https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/wss2.12065

2. **Passive Wi-Fi Probe-Request Localization**
   Sun et al. (2017), Mobile Information Systems.
   https://onlinelibrary.wiley.com/doi/10.1155/2017/7821585

3. **Arduino UNO Q — Product Documentation**
   https://www.arduino.cc/product-uno-q

4. **Qualcomm Arduino UNO Q Developer Resources**
   https://www.qualcomm.com/developer/hardware/arduino-uno-q

5. **Emergency Communication in Catastrophic Disasters**
   Procedia Computer Science (2017).
   https://www.sciencedirect.com/science/article/pii/S1877050917319993

6. **Random Forest / Ensemble Methods**
   https://scikit-learn.org/stable/modules/ensemble.html#forests-of-randomized-trees

7. **OpenStreetMap Mapping Data**

ADDITIONAL TECHNICAL RESOURCES

- **Arduino Docs** — https://ardupilot.org/
- **IEEE Xplore UAV Search & Rescue** — https://ieeexplore.ieee.org/
- **IEEE Xplore Wi-Fi RSSI Localization** — https://ieeexplore.ieee.org/
- **TensorFlow** — https://www.tensorflow.org/

UAV Search & Rescue https://ieeexplore.ieee.org/document/10493986/

Wi-Fi RSSI Localization https://ieeexplore.ieee.org/document/9606145/